28-feb

Computer Science & IT

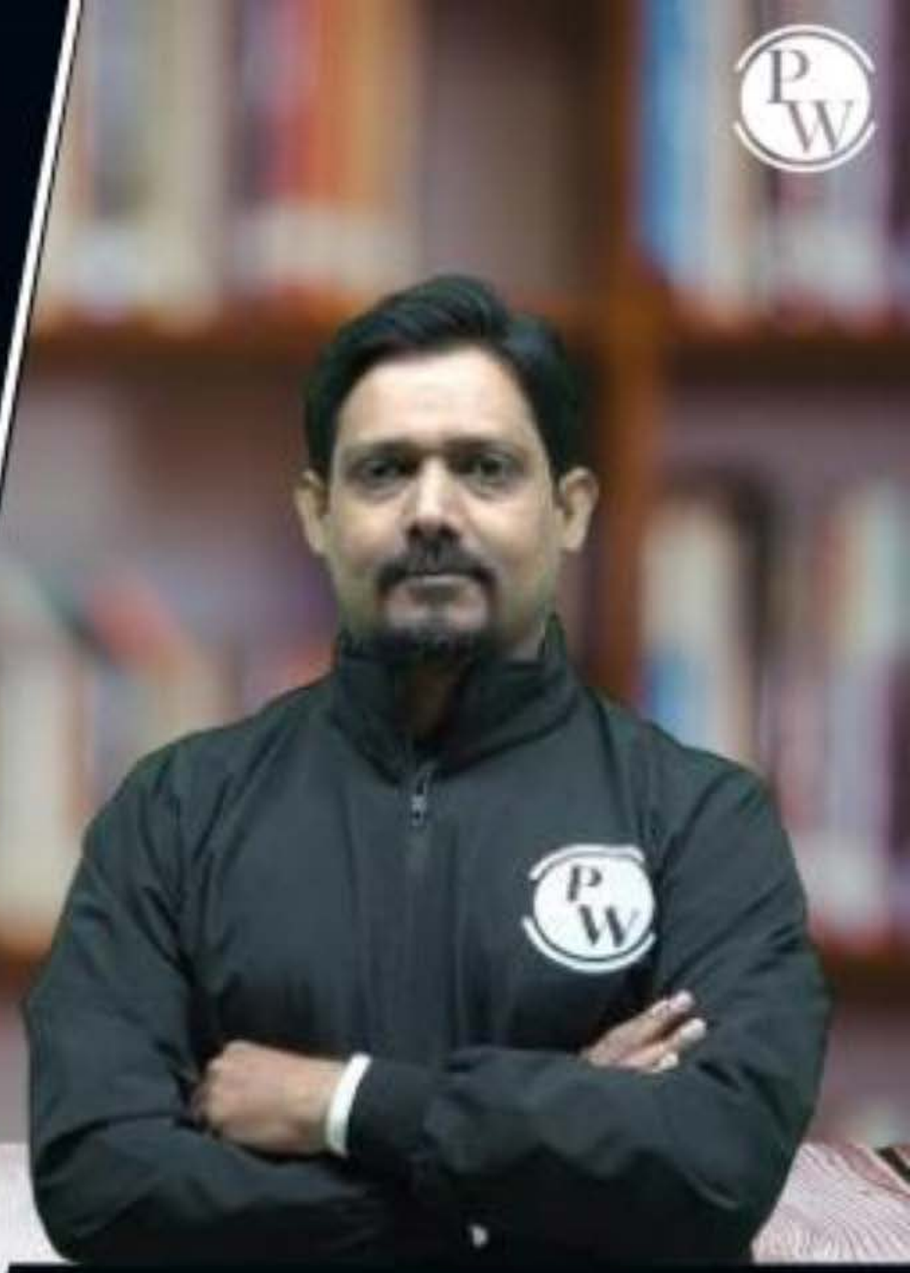
Score



Data Structure & Programming

Array 01

Lecture No. 01



By- Abhishek Sir

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Array

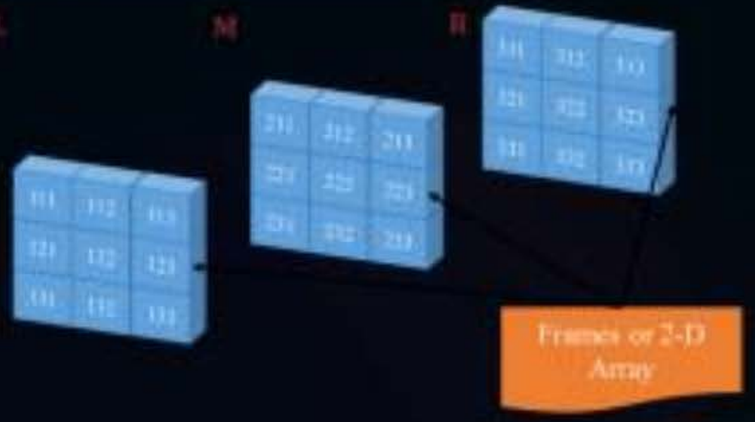
one dimension

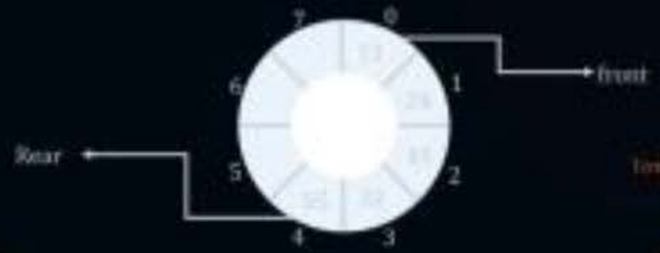
2-D Array

Slide

Topic : Data Structure

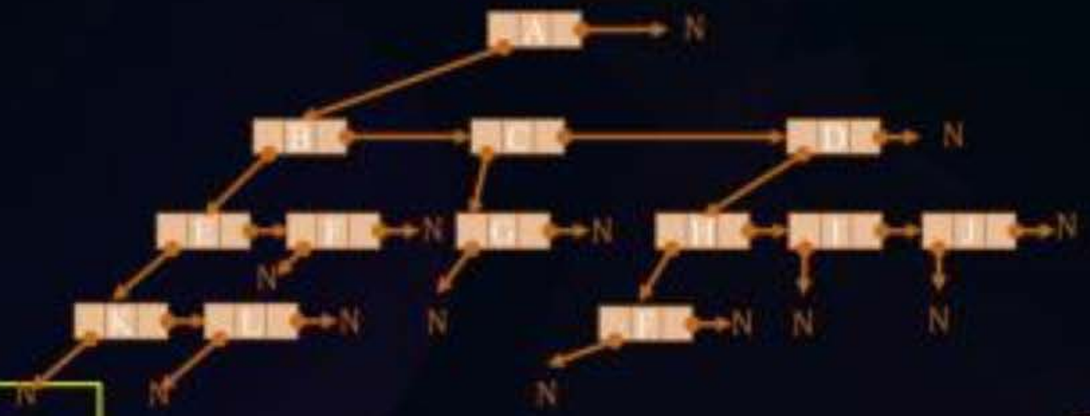


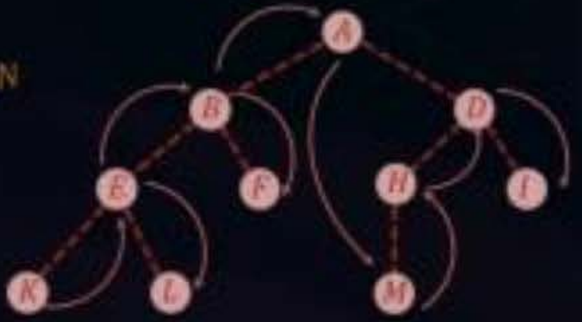














Slide



Topic : Syllabus

Linear Data structure

1. Array
2. stack
3. Queue
4. Linked List

Non Linear Data structure

1. Tree
2. Binary Search tree (AVL tree)
3. Binary Heaps
4. Hashing / Graph Representation

Slide



Telegram Group

@ABHISHEKSHARMAPW

Abhisheksharmapw

Slide

Topic : Weightage

10 – 11 – marks

6 – 7 Question



Slide



Topic : Definition

Collection of data

Data structure is about Data organization
Management, storage format that
enables effective Access & Modification.

} operation → Complexity of operation

Slide

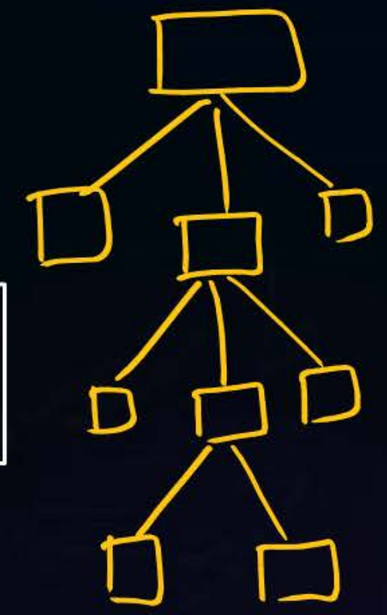


Topic : Array

int Size 4B

Array is Sequential collection of similar Similar Data type.

Sequential- Contiguous memory location





int a[10];

Address of A[9] = 100 + 8 X 4B
= 132

Slide



Topic : Array

A[1 .... 10]

A[10]

A[0 ... 9]

C - Lower bound always zero

Lower bound	upper bound
Index of first element	Index of Last element

Base Address :- Address of first element in Memory

Size of each element : Depends upon data type :-

Double - 8B
int - 4B
char - 1B
float - 4B

Slide

No of element in 1-D array UB - LB + 1

A[1 ... 100] :- 100 - 1 + 1 = 100

A[2 ... 73] 73 - 2 + 1 = 72

A[-5 ... 14] 14 − (−5) + 1 = 20



Topic : Question

#Q. Consider the above array A[1....100]. Base address of the array is 1000 and each element occupies 2 Bytes of space. What is the address of A[78]?



No of elements arranged - A[1] .. A[77] = 77 element

1000 + 77 x 2 = 1000 + 154 = 1154

Slide



Topic : Answer

We need the address of A[78], before reaching A[78] we have already arranged A[1.....77] each occupies 2B of space hence 77× 2 = 154B space , The final address is 1000+154 = 1154.



Topic : Question

#Q. Consider the above array A[0...99]. Base address of the array is 1000 and each element occupies 2 Bytes of space. What is the address of A[78].

Before A[78] No. of elements arranged A[0] .... A[77]

77 - 0 + 1 = 78

$1000 + 78 \times 2$

$= 1000 + 156 = 1156$

Slide



Topic : Question

Consider the above array A[0...99]. Base address of the array is 1000 and each element occupies 2 Bytes of space. What is the address of A[78].

$$1000 + (78 - 0) \times 2$$

$$= 1156$$

Slide



Topic : Question

#Q. Consider the above array A[-5.....100]. Base address of the array is 1000 and each element occupies 2-Bytes of space. What is the address of A[78].

-5 ...

No. of elements arranged

LB UB

-5 ... 77 = UB - LB + 1

77 - (-5) + 1

= 83

1000 + 83 × 2 = 1166

Slide



Topic : Question

#Q. Consider the above array A[-5.....100]. Base address of the array is 1000 and each element occupies 2-Bytes of space. What is the address of A[78].

-5 ...

$1000 + (78 - (-5)) \times 2$

$1000 + 83 \times 2 = 1166$

Slide



Topic : Solution

We need the address of A[78], before reaching A[78] we have already arranged A[-5 …..77] each occupies 2B of space hence 77-(-5) +1 = 83 elements already arranged. With each element occupy 2B total space is 83 × 2 = 166B. The final address is 1000+166 = 1166.



Topic :Generalized formula

A [LB ... UB]

BA - Base Address

S - Size

Address of A[i]

No. of elements arranged

A[LB] .... A[i-1]

UB - LB + 1

i - 1 - LB + 1

UB

(i - LB)

$$BA + (i - LB) \times S$$

BA + i x Size

C-Language

Lower bound 0

Slide



2- Dimensional Array

Topic : 2 D Array

	col-1	col-2	col-3	col-4
Raw-1	a_{11}	a_{12}	a_{13}	a_{14}
Raw-2	a_{21}	a_{22}	a_{23}	a_{24}
Raw-3	a_{31}	a_{32}	a_{33}	a_{34}
Raw4	a_{41}	a_{42}	a_{43}	a_{44}

a[1...4][1...4]

A[4][3]

Raw column

4th Raw 3rd column

Slide



Topic : 2 D Array

Row col
a[1..4][1..4]

	col-1	col-2	col-3	col-4
Row-1	a_{11}	a_{12}	a_{13}	a_{14}
Row-2	a_{21}	a_{22}	a_{23}	a_{24}
Row-3	a_{31}	a_{32}	a_{33}	a_{34}
Row4	a_{41}	a_{42}	a_{43}	a_{44}

No. of element in a Row :-
Size of column (No)

No. of element in a column
Size of row (No. of rows)

Slide

Topic : 2 D Array

	col-1	col-2	col-3	col-4
Raw-1	a_{11}	a_{12}	a_{13}	a_{14}
Raw-2	a_{21}	a_{22}	a_{23}	a_{24}
Raw-3	a_{31}	a_{32}	a_{33}	a_{34}
Raw4	a_{41}	a_{42}	a_{43}	a_{44}

⟹ 1-D array mapping

1. Raw major order (C-Language)
2. Column major order

Raw major order

a_{11}	a_{12}	a_{13}	a_{14}	a_{21}	a_{22}	a_{23}	a_{24}	a_{31}	a_{32}	a_{33}	a_{34}	a_{41}	a_{42}	a_{43}	a_{44}

Slide

Topic : 2 D Array

a[1..4][1..4]

	col-1	col-2	col-3	col-4
Row-1	a_{11}	a_{12}	a_{13}	a_{14}
Row-2	a_{21}	a_{22}	a_{23}	a_{24}
Row-3	a_{31}	a_{32}	a_{33}	a_{34}
Row4	a_{41}	a_{42}	a_{43}	a_{44}

a_{43} → 4th Row column

No. of Rows completely arranged $(4-1)=3$

No. of element in each Row :- 4

total : $3 \times 4 = 12$

---

in 4th Row, we are in 3rd column :

Before 3rd colum No. of element arranged :

$3-1=2$

Row major order

a_{11}	a_{12}	a_{13}	a_{14}	a_{21}	a_{22}	a_{23}	a_{24}	a_{31}	a_{32}	a_{33}	a_{34}	a_{41}	a_{42}	a_{43}	a_{44}
100	102	104	106	108	110	112	114	116	118	120	122	124	126	128	130

Slide



Topic : 2 D Array

	col-1	col-2	col-3	col-4
Raw-1	a_{11}	a_{12}	a_{13}	a_{14}
Raw-2	a_{21}	a_{22}	a_{23}	a_{24}
Raw-3	a_{31}	a_{32}	a_{33}	a_{34}
Raw4	a_{41}	a_{42}	a_{43}	a_{44}

Slide

Topic : 2 D Array

	col-1	col-2	col-3	col-4
Raw-1	a_{11}	a_{12}	a_{13}	a_{14}
Raw-2	a_{21}	a_{22}	a_{23}	a_{24}
Raw-3	a_{31}	a_{32}	a_{33}	a_{34}
Raw4	a_{41}	a_{42}	a_{43}	a_{44}

total = $3 \times 4 = 12$ – Raw

$3 - 1 = 2$

14

$100 + 14 \times 2 = 128$

Slide



Topic : Question

#Q. Consider the following array A[1...100][1...100].Base address is 1000 and each element occupies 2-Bytes of space. What is the address of A[50][49] in row major order?

A[50][49]

Before 50th Raw : No. of Raws completly arranged

(50 - 1) = 49

No. of element in each Raw : 100 total : 49x100

4900

50th Raw, No. of element arranged : (49 - 1) = 48

1000 + (4900 + 48) = 1000 + 4948x2 = 9896 = 10896

1000

Slide



Topic : Question

We need to find the address of A[50][49]. Before reaching to 50th row 1...49 rows are already been arranged and each row consists of 100 elements. So total 49× 100 =4900 elements are arranged.
For the 50th row we have already arranged 1...48 i.e. 48 elements (look at the second dimension) are already arranged.
So total number of elements arranged are = 4900+48 = 4948.
The address is 1000+4948× 2=10896

Slide



Topic : 2 D Array

A [20][50] declared A [1...20][1...50]

BA - 100

Size = 2B

Address : A[15][31]

No. of Rows arranged (15-1) = 14

No. of element in each Row 50

total = 14 x 50 = 700

on 15th Row 31st column

No. of elements (31-1) = 30

100 + (700+30) x 2 = 100 + 730 x 2

100 + 1460 = 1560

Slide



Topic : 2 D Array

$A[-2...10][7..20]$

LB_2 UB_2

Base - 100, Size 2B

Add. $A[7][10]$

Before 7th Row No. of Rows arranged

$(7-(-2)) = 9$

No. of elements in Row : $\underset{2}{UB} - \underset{2}{LB} + 1 = (20-7+1)$

total : $9 \times 14 = 126$ 14

In 7th Row : 10th column

No. of column arranged : $(10-7) = 3$

total = $126+3 = 129$

$100 + 129 \times 2 = 358$

Slide



2 mins Summary

Topic

1-D array

Topic

2-D - Row major order

Topic

UB - LB + 1 No. of

Topic

1D = BA + (i - LB) x Size

Topic

Slide

THANK - YOU